2ND SEMESTER LAB REPORT

TITLE : Electrical Circuit Analysis Lab



Faiq Ali (249520)

Muhammad Zeeshan (243346)

Muhammad Waqas (240756)

BACHELOR OF COMPUTER ENGINEERING (Fall-2025)

Submitted To:

Engr. Shanzay Jamil

Lab Engineer

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING FACULTY OF ENGINEERING AIR UNIVERSITY, ISLAMABAD

	AIR UNIVERSITY
	DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING
	EXPERIMENT NO 11

Lab Title: IMPLEMENTATION AND VERIFICATION OF NORTON THEOREM BOTH ON HARDWARE AND SOFTWARE

Objectiv: NORTON THEOREM

Student Name: Muhammad Waqas -Muhammad Zeesha -Faiq Ali

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks:

Date: 8/May/2025 Signature:

LABORATORY EXPERIMENT NO. 11

IMPLEMENTATION AND VERIFICATION OF NORTON THEOREM BOTH ON HARDWARE AND SOFTWARE

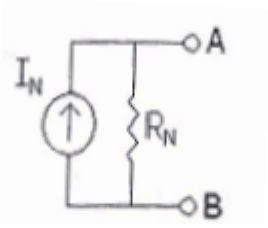
OBJECTIVE:

1. To find experimentally the values of components required to construct a Norton's equivalent circuit.
2. To check the experimental values versus calculated values.
3. To check the validity of Norton's theorem.
4. To verify the results through software using Proteus.

Norton's Theorem

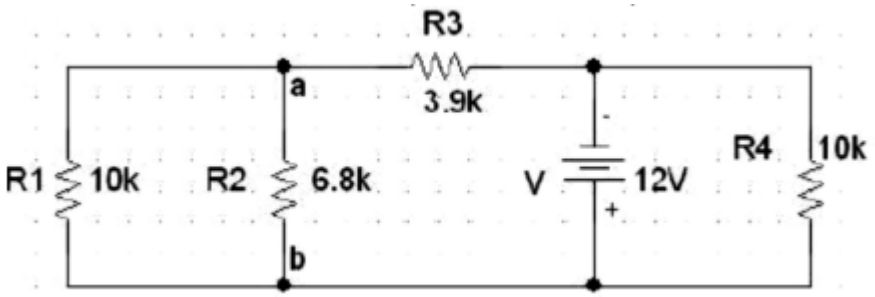
This lab deals with another network theorem known as the "Norton's Theorem".
Norton's theorem as the Thévenin's Theorem, allows one to replace a complicated circuit with a simple equivalent circuit containing only a voltage source and a series connected resistor. The theorem is very important from both theoretical and practical viewpoints. Concisely stated, Norton's Theorem says:

"Any two-terminal linear circuit can be replaced by an equivalent circuit consisting of a current source (I_N) and a parallel resistor (R_N)"



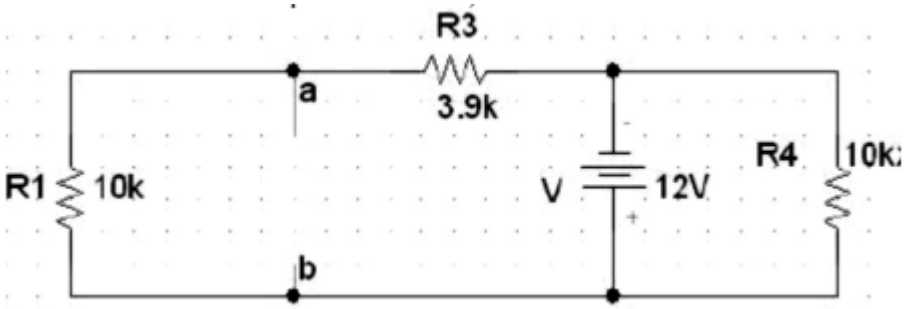
Lab Tasks:

Lab task 1:



d) Measure the value of V L and I L using a DMM. Record these values and compare them with the theoretical values of V L and I L.

VL = 6.11V **IL = 0.90mA**



f) Turn on the trainer and using a DMM, measure the voltage across the terminals A and B i.e. the open-circuit voltage Voc or Vth. Record this value and compare it with the theoretical value of Vth.

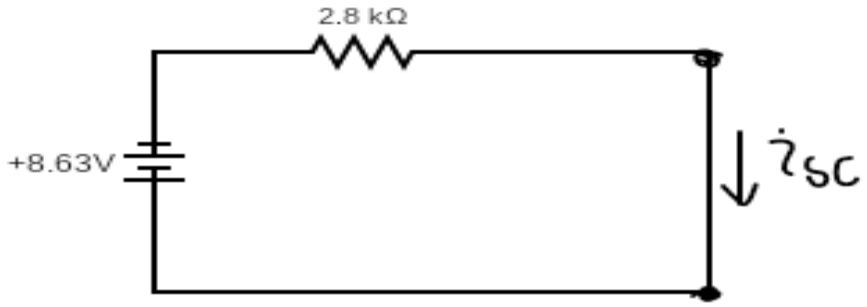
Vth = 8.63

Also measure the short-circuit current, I sc by placing the DMM directly between terminals A and B, as shown. Record this value and compare it with the theoretical value of I sc.

Isc = 3.09mA

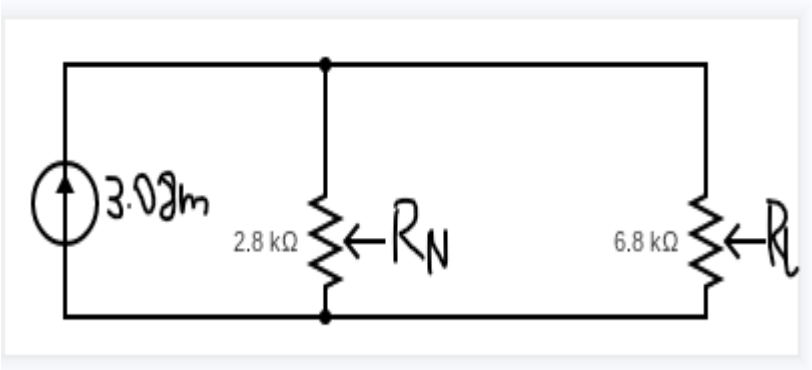
I) Turn on the trainer and measure the short circuit current I sc between terminals A and B. this value should be the same as the one calculated in step (g) and also the same as the theoretical value of Isc.

Isc = 3.08



n) Turn on the trainer and measure V L , the voltage across load, as Valus are totally same to the original circuit

VL =<u>6.11 V</u> **IL = <u>0.90mA</u>**



Theoretical Calculation

Thévenin Analysis

Step 1: Calculate VTH

VN = Va – Vb

Va = 12V × 10kΩ/(3.9kΩ + 10kΩ) = 12V × 10/13.9 = 8.63V

Vb = 0V

VN = 8.63V – 0

Norton VN = 8.63V

Step 2: Calculate RN

RN = Rth = R1 || R3

R1 = 3.9kΩ

R3 = 10kΩ

RN = (R1 * R3) / (R1 + R3)

RN = (3.9k * 10k) / (3.9k + 10k)

RN = 39k^2 / 13.9k

RN = 39k / 13.9

RN ≈ 2.8058 kΩ

Step 3: Calculate Voltage Across R2

I = VTH/(RTH + R2) = 8.63V/(12.81kΩ + 6.8kΩ)

= 8.63V/19.61kΩ = 0.44mA VR2 = I × R2 = 0.44mA × 6.8kΩ = 2.99V

Corrected Analysis

VTH = 8.63V I = 8.63V/RTH = 8.63V/12.81kΩ

= 0.674mA VR2 = 8.63V × 10kΩ/(10kΩ + 6.8kΩ)

= 8.63V × 10/16.8 = 5.14V

Nodal Analysis

(Va-12V)/3.9kΩ + (Va-Vb)/6.8kΩ = 0

(Vb-Va)/6.8kΩ + Vb/10kΩ = 0

Vb = Va × 10kΩ/(10kΩ+6.8kΩ) = 0.595

Va (Va-12V)/3.9kΩ + (Va-0.595Va)/6.8kΩ = 0

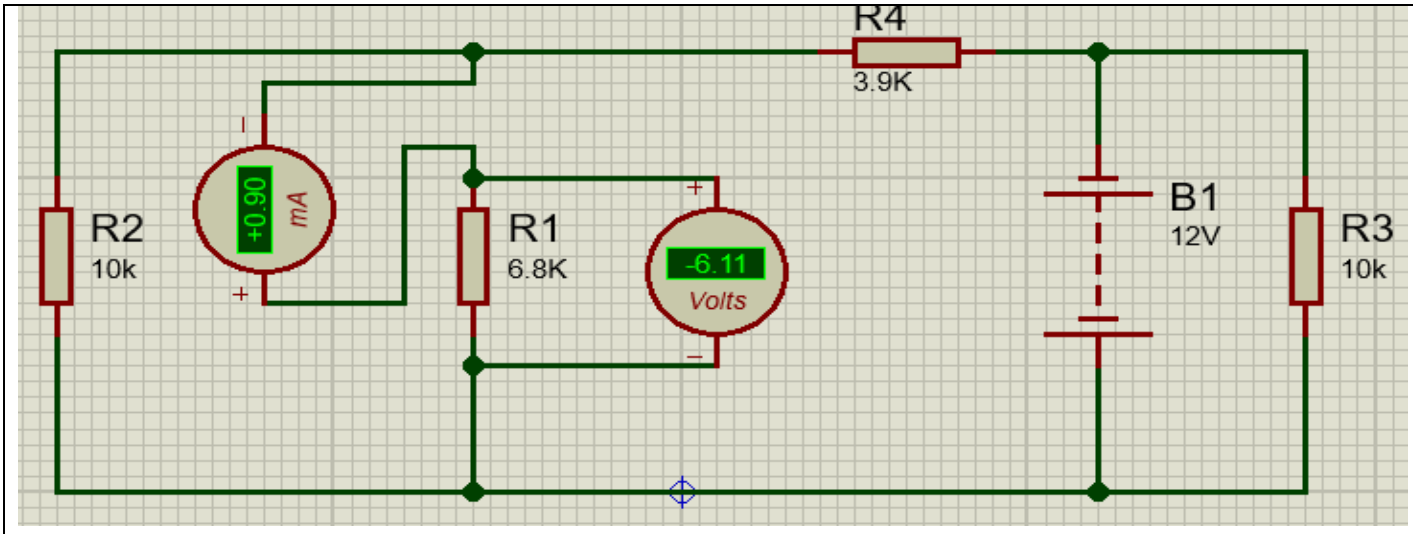
(Va-12V)/3.9kΩ + 0.405Va/6.8kΩ = 0

Va = 10.74V Vb = 0.595 × 10.74V = 6.39V

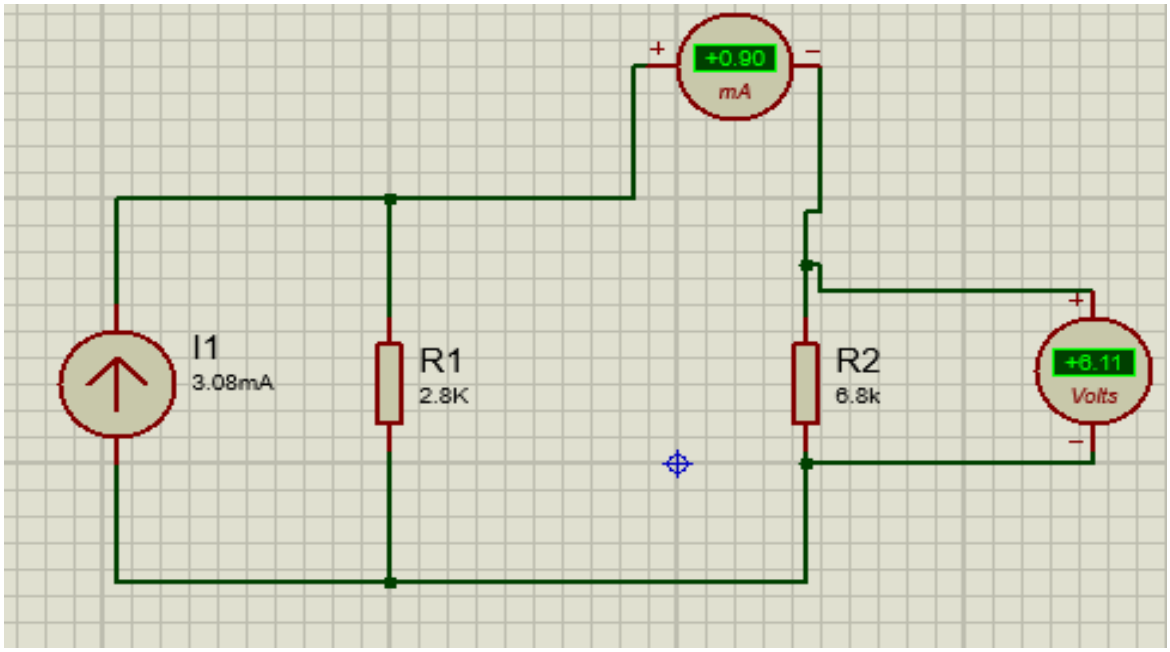
VR2 = Va - Vb = 10.74V - 4.39V = 6.35V

Simulation result: VR2 = 6.11V

Protius Simulation

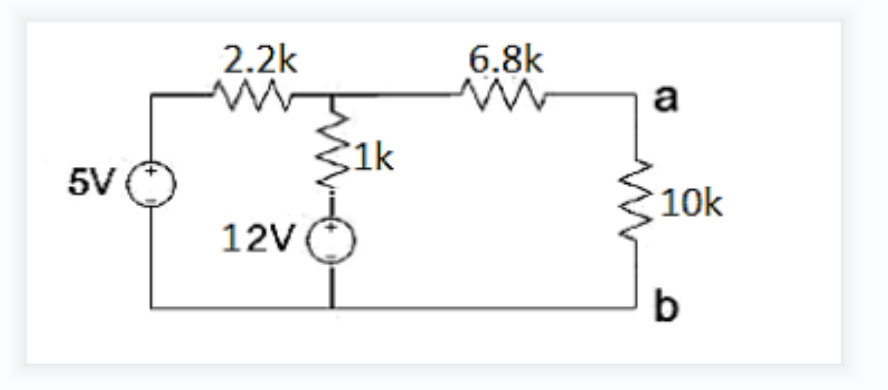


NORTON'S CIRCUIT:

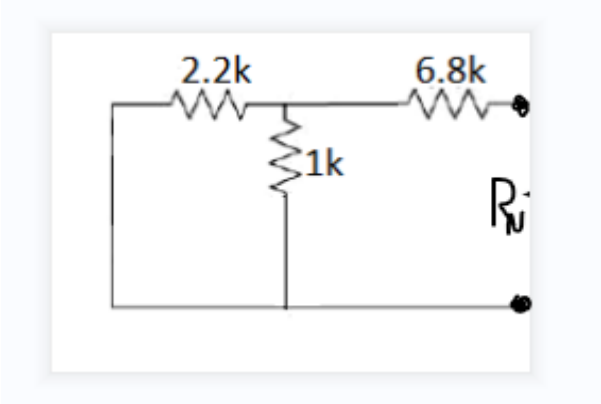


Lab Task 2:

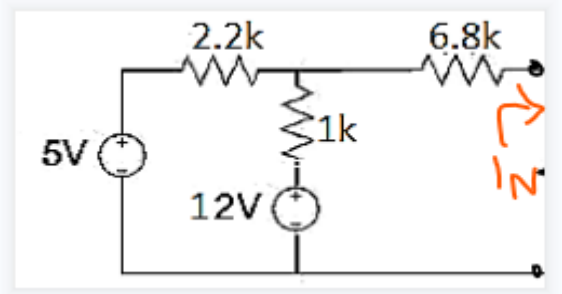
CIRCUIT DIAGRAM



The portion of the circuit outlined is to be replaced by it Norton equivalent circuit.
To determine R N, set all sources to zero by replacing voltage source by short-circuit and current source by an open circuit, as shown:



To determine IN, the sources are replaced and the open-circuit voltage is determined as shown:



5 = (2I1 + (1k)I1 + 12 (3.2k)I1 = -7 I1 = -7 / (3.2k) = -2.2mA

V1k = I1 × 1k

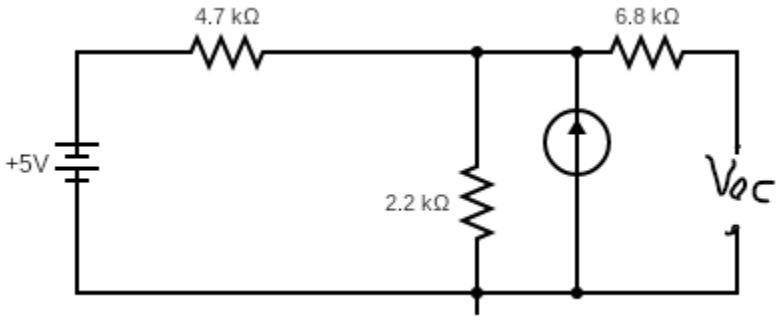
V1k = -2.2m × 1k = -2.2V

VN = 12V + (-2.2V) = 9.8V

IN = VN / RN

IN = 9.8 / 7.5k = 1.31mA

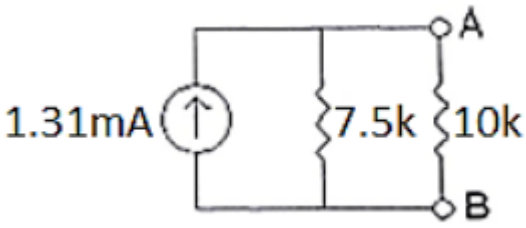
Thévenin Equivalent OLTAGE:



(v - 12) / 4700 + v / 2200 = 3

VTH ≈ 4499.48

Norton Equivalent Circuit:



$$\mathbf{I_L = \frac{R_N}{R_N + R_L} \times I_N}$$

IL = (7.5k / (7.5k + 10k)) × 1.31m = 0.56mA

VL = IL × RL = 0.56m × 10k = 5.6V

LOAD VOLTAGE = 5.6V

LOAD CURRENT =0.56mA

Theoretical Calculation

NORTON RESISTANCE(RN)

Now, solving for RN:

(2.2k || 1k) + 6.8k = 7.5kΩ

2.2k || 1k = (2.2k × 1k) / (2.2k + 1k)

2.2k || 1k = 2200 / 3200

2.2k || 1k = 0.6875k

2.2k || 1k = 687.5Ω

RN = 687.5Ω + 6.8k

RN = 687.5Ω + 6800Ω

RN = 7487.5Ω

RN ≈ 7.5kΩ

Norton Current (IN)

5 = (2.2k)I1 + (1k)I1 + 12 (3.2k)I1 = -7 I1 = -7 / (3.2k) = -2.2mA

V1k = I1 × 1k

V1k = -2.2m × 1k = -2.2V

VN = 12V + (-2.2V) = 9.8V

IN = VN / RN

IN = 9.8 / 7.5k = 1.31mA

Verification with 10kΩ load

IL = (7.5k / (7.5k + 10k)) × 1.31m = 0.56mA

VL = IL × RL = 0.56m × 10k = 5.6V

LOAD CURRENT

I_L ≈ 0.56mA

LOAD VOLTAGE

VL = IL × RL

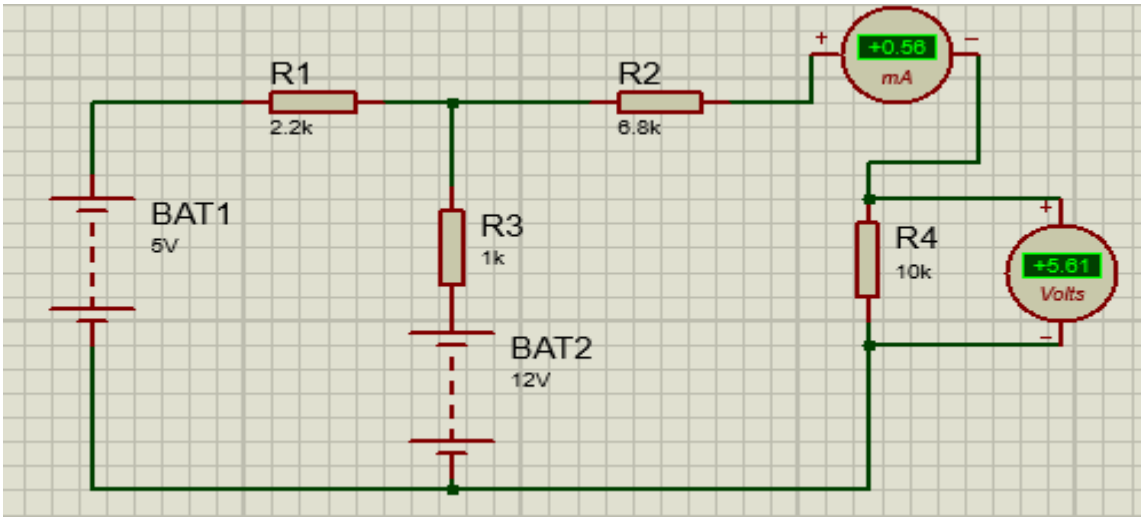
VL = 0.2459 × 10000

VL = 2459 V

V_L ≈= 5.6V

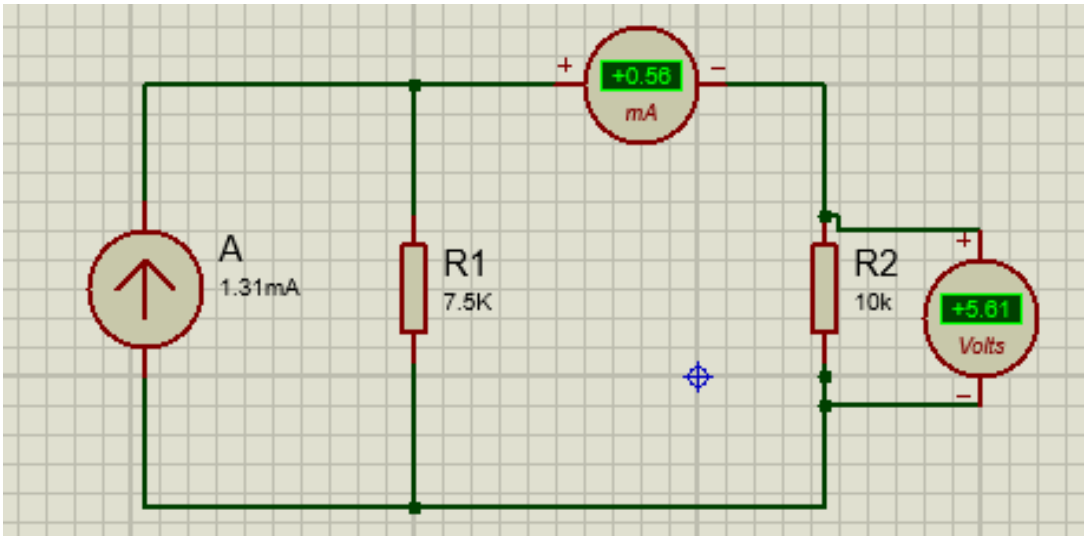
Protius Simulation

Original Circuit



NORTON Equivalent Circuit:

As values are same in original circuit and in final Thévenin Equivalent Circuit:



Conclusions:

Norton's Theorem is a powerful tool that makes complex circuits feel simple. It tells us that any network of resistors and sources, as seen from two terminals, can be replaced by just one current source in parallel with one resistor. To build this equivalent, you first "turn off" all voltage and current sources to find the resistance RNR_N, then you measure the short-circuit current INI_N between the terminals. Once you know INI_N and RNR_N, you can swap the entire original network for a neat little package—a current source of INI_N flowing into a resistor RNR_N. This shortcut saves you time and effort whenever you want to see how different loads behave without reworking the whole circuit.